8	**(a)**		$[\frac{19.8}{36}$ =] 0.55	**B1** **[1]**	**1.1**	or $\frac{11}{20}$	Must be dec or simplified frac; but condone 55%
8	**(b)**		0.0380(96943116…) BC	**B1** **[1]**	**1.1**	Correct to at least 2 sf	Allow truncated

8	**(c)**		H_0: p = 0.45 H_1: p < 0.45	**B1**	**1.1**	OR H_0: p = 0.55 H_1: p > 0.55	Both hypotheses
			p is the probability that a driver speeding / caught speeding / on speed awareness course is female	**B1$**	**2.5**	p is the probability that a driver speeding / caught speeding / on speed awareness course is male	Must not be inconsistent with H_1
			Use or sight of B(25, 0.45)	**M1***	**3.3**	OR Use or sight of B(25, 0.55)	Can be implied by P(X = 7) (OR P(X = 18)) = 0.038..
			P($X \leq 7$) = 0.06385(0773831)	**A1**	**1.1**	P($X \geq 18$) = 1 – P($X \leq 17$) = 1 – 0.936149 = 0.06385	
			0.06385 > 0.05	***M1**	**3.4**	FT their probability	
			Result is NOT significant or "accept H_0"	**B1**	**1.1**	Dep on 0.06385	May be incorporated in conclusion
			There is insufficient evidence (at the 5% level) to suggest that drivers speeding / caught speeding / on speed awareness course are more likely to be male. oe	**E1** **[7]**	**2.2b**	Dep all previous marks except B1$ SC: Use of p = 0.5 can score B1B1M1 (can be implied by 0.0216 or 0.0143) A0M1B0E0	Not assertive
8	**(d)**		eg Each driver speeding / caught speeding has the same (constant) probability of being selected for the speed awareness course.	**E1** **[1]**	**2.4**	OR the probability of any particular driver caught speeding being selected for the speed awareness course is independent of any other driver caught speeding being selected.	